

Phishing Email Analysis

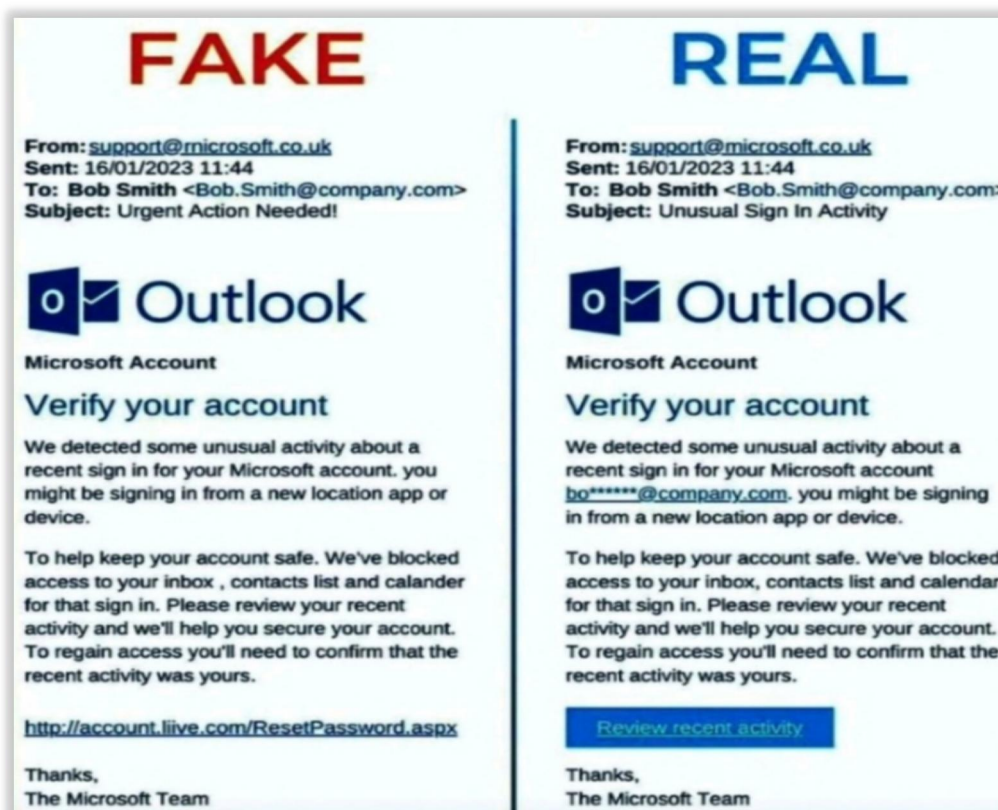
Learning Summary:

I gained hands-on experience in identifying and analyzing suspicious emails. I learned how to extract and interpret email headers, verify sender authentication using SPF, DKIM, and DMARC, and trace the email path using hops. I also used tools like VirusTotal, AbuseIPDB and MxToolbox to detect phishing indicators and perform IP and domain reputation checks. This process enhanced my practical skills in threat identification, email forensics, and cybersecurity reporting.

Tasks :

Task 1: Phishing email:

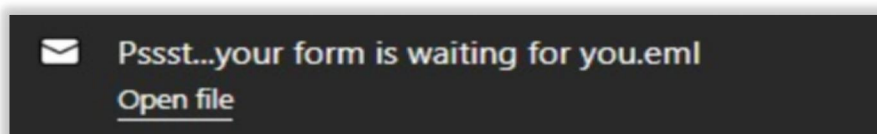
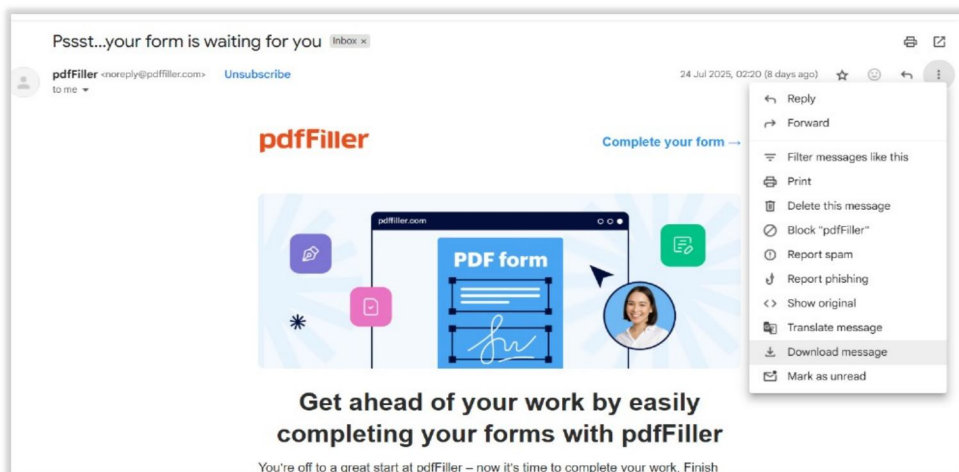
Phishing is a form of online fraud in which hackers attempt to get your private information such as passwords, credit cards, or bank account data.



Source: LinkedIn

Task 2: Download an Email (.eml format) from email client:

Email Sender: noreply@pdffiller.com



Task 3: Email Protocols (SMTP, POP/IMAP):

Email Protocols:

Email protocols are sets of rules that govern how emails are sent, received, and stored.

The main protocols are:

1. SMTP: Stands for Simple Mail Transfer Protocol, and it is responsible for sending email messages. This protocol is used by email clients and mail servers to exchange emails between computers.

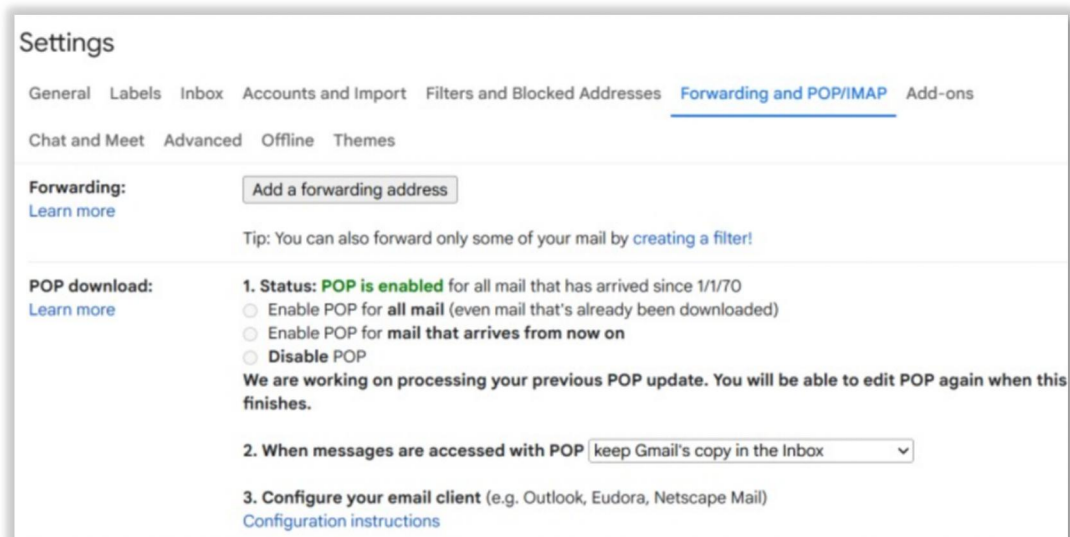
Working: A mail client and the SMTP server communicate with each other over a connection established through a particular email port. Both entities are using SMTP commands and replies to process your outgoing emails.

Example: noreply@pdffiller.com

```
Delivered-To: mehrabshaheen.13@gmail.com
Received: by 2002:a05:6504:33c6:b0:2bb:e908:fbcc with SMTP id a6csp511731lty;
  Wed, 23 Jul 2025 14:20:12 -0700 (PDT)
X-Google-Smtp-Source: AGHT+IEkH/wznPGuyYw23ik1UNUX24uov0ECQgefRvngMsqLHZc48drneHXLxpufPTCrohIQSwmx
X-Received: by 2002:a05:620a:844:b0:7d4:49fa:3c59 with SMTP id af79cd13be357-7e62a0e4fc4mr559626485a.15.1753305611914;
  Wed, 23 Jul 2025 14:20:11 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1753305611; cv=none;
  d=google.com; s=arc-20240605;
  b=j/+eguJ4PnAG5R16WRdf7+Ai5Mk9/W6G3DkeyPkegl0tlV4EPtkaB8GmErgvmxfie
  3QHkTCJ19VvQlsAZUBChWcPPm+3tsiModbGK+L9ymilk/Vu50oeBYL6vGm/XY316sjpf
  +WcDIrqTfrihs3GKthj+DyAXD4IrFV17js8hAbVsrue7FkhYzvPagSr07dCXrcGumbr
  SZ5XK0i49EM6ekZ8H8eG6TKLvsdZiH8uTvo8uKZwBSZYswMTWxrlLuhdEclTea2NRe
```

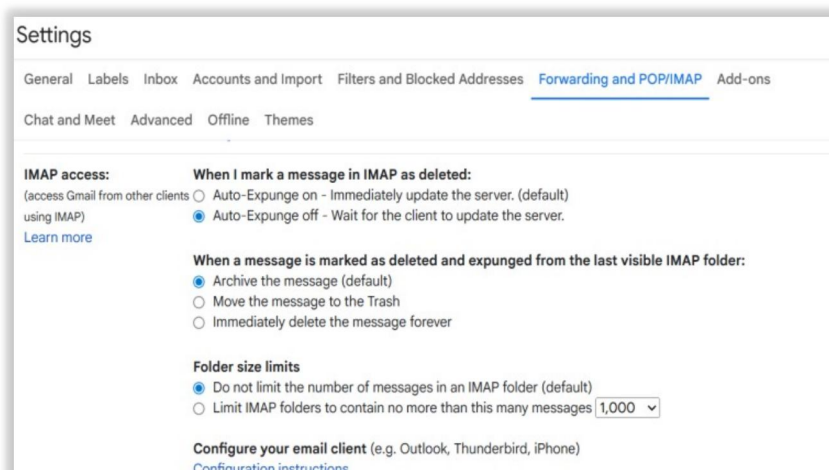
2. POP3: Stands for **Post Office Protocol version 3**, which provides access to an inbox stored in an email server. It executes the download and deletes operations for messages.

Working: Thus, when a POP3 client connects to the mail server, it retrieves all messages from the mailbox. Then it stores them on your local computer and deletes them from the remote server.



2. IMAP: The Internet Message Access Protocol (IMAP) allows you to access and manage your email messages on the email server.

Working: This protocol permits you to manipulate folders, permanently delete and efficiently search through messages. It also gives you the option to set or remove email flags, or fetch email attributes selectively. By default, all messages remain on the server until the user specifically deletes them.



Task 4: Analyze phishing email manually:

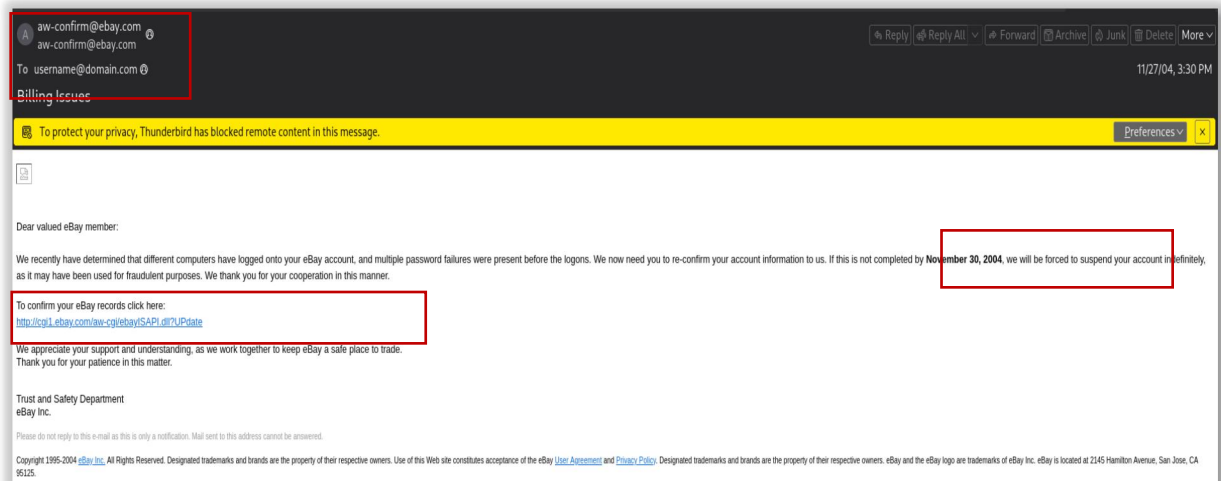
Phishing Email Sample Source: monkey.org

Opened in thunderbird

Email Sender: aw-confirm@ebay.com

Index of /~jose/phishing			
Name	Last modified	Size	Description
Parent Directory		-	
20051114.mbox	2014-02-10 17:00	3.9M	
README.txt	2018-07-13 17:15	844	
phishing-2013	2018-05-15 19:18	8.4M	
phishing-2016	2018-05-15 19:18	12M	
phishing-2017	2018-05-15 19:18	11M	
phishing-2018	2019-01-11 19:41	8.4M	
phishing-2019	2020-01-03 15:08	5.0M	
phishing-2020	2021-03-08 15:59	5.6M	
phishing-2021	2022-02-12 16:02	2.6M	
phishing-2022	2023-01-31 21:15	4.6M	
phishing-2023	2024-01-01 16:31	7.4M	
phishing-2024	2025-01-05 18:32	15M	
phishing0.mbox	2005-06-13 14:37	3.0M	
phishing1.mbox	2005-11-15 16:01	4.0M	
phishing2.mbox	2006-08-07 13:09	9.9M	
phishing3.mbox	2007-08-07 01:20	19M	
private-phishing4.mbox	2018-07-24 21:46	31M	

Apache/2.4.62 (Debian) Server at monkey.org Port 443



Email header of Sample Email

```
From: aw-confirm@ebay.com Sat Nov 27 15:33:27 2004
Return-Path: <aw-confirm@ebay.com>
X-Original-To: username@domain.com
Delivered-To: username@domain.com
Received: from 81.18.87.130 (unknown [81.18.87.130])
    by spanky.domain.com (Postfix) with SMTP id CCC115378FC
    for <username@domain.com>; Sat, 27 Nov 2004 15:33:24 -0500 (EST)
Received: from 20.84.152.113 by 65.23.81.142; Sat, 27 Nov 2004 15:24:27 -0500
Message-ID: <LCCUVYFMKBWQCHUDWYUCUYU@notmail.com>
From: "aw-confirm@ebay.com" <aw-confirm@ebay.com>
Reply-To: "aw-confirm@ebay.com" <aw-confirm@ebay.com>
To: username@domain.com
Subject: Billing Issues
Date: Sun, 28 Nov 2004 00:30:27 +0400
MIME-Version: 1.0
Content-Type: multipart/alternative;
    boundary="--591699981497957"
X-Priority: 3
X-CS-IP: 224.248.218.116
Status: 0
X-Status:
X-Keywords:
X-UID: 1
```

- Check Email Header.
- Count number of Hops and check their IPs, unknown IPs mean Suspicious Email.
- Check SPF, DKIM, DMARC policies if they are missing or not passed then the email might be fake.
- Phishing Email use urgent messages like 'Billing Issue'.
- Check spelling and grammar mistakes.
- Hover on link given in email.

Note: SPF, DKIM, DMARC policies are missing in Phishing Email Sample.

Task 5. Total hops covered before receiving the email in destination

Email Sender: aw-confirm@ebay.com

Email header of Sample Email:

- Open raw header of phishing email sample.
- Look for all lines starting with Received.

```
From: aw-confirm@ebay.com Sat Nov 27 15:33:27 2004
Return-Path: <aw-confirm@ebay.com>
X-Original-To: username@domain.com
Delivered-To: username@domain.com
Received: from 81.18.87.130 (unknown [81.18.87.130])
    by spanky.domain.com (Postfix) with SMTP id CCC115378FC
    for <username@domain.com>; Sat, 27 Nov 2004 15:33:24 -0500 (EST)
Received: from 20.84.152.113 by 65.23.81.142; Sat, 27 Nov 2004 15:24:27 -0500
Message-ID: <LCCOVYFMRBWOCHUDWYCOYUW@hotmail.com>
From: "aw-confirm@ebay.com" <aw-confirm@ebay.com>
Reply-To: "aw-confirm@ebay.com" <aw-confirm@ebay.com>
To: username@domain.com
Subject: Billing Issues
Date: Sun, 28 Nov 2004 00:30:27 +0400
MIME-Version: 1.0
Content-Type: multipart/alternative;
    boundary="-591699981497957"
X-Priority: 3
X-CS-IP: 224.248.218.116
Status: 0
X-Status:
X-Keywords:
X-UID: 1
```

Result:

Hops Numbers are 2.

Task 6. Email policies (SPF, DMARC etc):

DMARC, DKIM, and SPF are three email authentication methods. Together, they help prevent spammers, phishers, and other unauthorized parties from sending emails on behalf of a domain* they do not own.

1. Sender Policy Framework (SPF):

- Allows a domain to list all authorized email sending servers.
- SPF records contain the IP addresses of permitted servers. Receiving mail servers check incoming emails against this record to verify legitimacy before delivery.

2. Domain Keys Identified Mail (DKIM):

- Enables domain owners to digitally "sign" emails from their domain, confirming their origin.
- It uses public key cryptography. The sender signs the email with a secret private key, while the public key is stored in the domain's DKIM record. Receiving servers use this public key to verify the signature.

3. Domain-based Message Authentication Reporting and Conformance (DMARC):

- DMARC policies in DMARC records instruct mail servers to reject emails that fail SPF or DKIM.

Content Source: siteground

Email Sender: noreply@pdffiller.com (used in Q2)

Message ID	<17533056107529.1.29519.5177630516@pdfiller.com>
Created on:	24 July 2025 at 02:20 (Delivered after 1 second)
From:	pdfFiller <noreply@pdfiller.com>
To:	"mehrabshaheen.13@gmail.com" <mehrabshaheen.13@gmail.com>
Subject:	Pssst...your form is waiting for you
SPF:	PASS with IP 34.227.182.133 Learn more
DKIM:	'PASS' with domain pdfiller.com Learn more
DMARC:	'PASS' Learn more

```

data=google.com
ARC-Authentication-Results: i=1; mx.google.com;
dkim=pass header.i=@pdfiller.com header.s=ae1 header.b="PGLr9QV/";
spf=pass (google.com: domain of noreply@pdfiller.com designates 34.227.182.133 as permitted sender) smtp.mailfrom=noreply@pdfiller.com;
dmarc=pass (p=REJECT sp=REJECT dis=NONE) header.from=pdfiller.com
Return-Path: <noreply@pdfiller.com>
Received: from smtp4-7.pdfiller.com (smtp4-7.pdfiller.com. [34.227.182.133])
    by mx.google.com with ESMTPS id a79cd13be357-7e35ec8dd29s1841316485a.773.2025.07.23.14.20.11
    for <mehrabshaheen.13@gmail.com>
    (version=TLS1_3 cipher=TLS_AES_256_GCM_SHA384 bits=256/256);
    Wed, 23 Jul 2025 14:20:11 -0700 (PDT)
Received-SPF: pass (google.com: domain of noreply@pdfiller.com designates 34.227.182.133 as permitted sender) client-ip=34.227.182.133;
Authentication-Results: mx.google.com;
dkim=pass header.i=@pdfiller.com header.s=ae1 header.b="PGLr9QV/";
spf=pass (google.com: domain of noreply@pdfiller.com designates 34.227.182.133 as permitted sender) smtp.mailfrom=noreply@pdfiller.com;
dmarc=pass (p=REJECT sp=REJECT dis=NONE) header.from=pdfiller.com

```



```
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/simple; d=pdffiller.com; s=ae1; t=1753305611; bh=LkeSm8+31SG70KT+nrQk+CKQ31kZmDGSnvJ90cIw4IM=;
h=Date:Subject:From:To:List-Unsubscribe:List-id:From; b=PGLr9QV/f05FC17TKJ9Z/+V7XlNU3XogfDyfvNRqGL4PvNXuqqqZP+P4qRRwdZ9uB
NYDSysTnmpAGhKxyHbChxetMlxH8Bc80hRSxCzmcEXw0S8gKrQJEdf3StoGpv37g3
kzFN+WE2sUjby0KXm+gijlil/eAPAz964FGLiBgR01aYz7OzQH0XKYapDYBrS3pnX
Yrtix308W0Gxucgaxs1K0Pp154drxmLwutHMS5srFAfRcmVL+2bvegse3cp6Pvz0waJ
1W+EJYKJ00ATThgDtoMKNZw90iJLKyxu6LXEvcYOe+fb+QxEPcQFNngzC7XqulZjzq
nupya67o5rRfA==
```

Result:

In given Email email policies are passed that shows email is valid.

Content Source: cloudfare

Task 7. Confirm if the email is spoofed

Email Sender: aw-confirm@ebay.com

Email header of Sample Email:

```
From: aw-confirm@ebay.com Sat Nov 27 15:33:27 2004
Return-Path: <aw-confirm@ebay.com>
X-Original-To: username@domain.com
Delivered-To: username@domain.com
Received: from 81.18.87.130 (unknown [81.18.87.130])
    by spanky.domain.com (Postfix) with SMTP id CCC115378FC
    for <username@domain.com>; Sat, 27 Nov 2004 15:33:24 -0500 (EST)
Received: from 20.84.152.113 by 65.23.81.142; Sat, 27 Nov 2004 15:24:27 -0500
Message-ID: <LCCOVYFMRBQCHUDWYCOYUW@hotmail.com>
From: "aw-confirm@ebay.com" <aw-confirm@ebay.com>
Reply-To: "aw-confirm@ebay.com" <aw-confirm@ebay.com>
To: username@domain.com
Subject: Billing Issues
Date: Sun, 28 Nov 2004 00:30:27 +0400
MIME-Version: 1.0
Content-Type: multipart/alternative;
    boundary="--591699981497957"
X-Priority: 3
X-CS-IP: 224.248.218.116
Status: 0
X-Status:
X-Keywords:
X-UID: 1
```

Result:

- Mismatch between 'From' Address and Message Source (Message Id should come from ebay.com not hotmail.com).
- Missing authentication Mechanisms(SPF, DKIM, DMARC).
- IPs do not belong to eBay's official mail servers.
- Date and time mismatch it shows header was tampered.

Tool Used: AbuseIPDB

Phishing Email Sample: (already used in Q4)

Email Sender: aw-confirm@ebay.com

Verification of Hop's IPs

Check an IP Address, Domain Name, or Subnet
e.g. 2404:3100:1007:c789:851e:188f:1eb5:3b62, microsoft.com, or 5.188.10.0/24

81.18.87.190 CHECK

81.18.87.190 was not found in our database

ISP	RCS & RDS Residential
Usage Type	Fixed Line ISP
ASN	Unknown
Hostname(s)	static-81-18-87-190.severin.rdsnet.ro
Domain Name	digi.ro
Country	 Romania
City	Giurgiu, Giurgiu County

AbuseIPDB » **81.18.87.130**

Check an IP Address, Domain Name, or Subnet
e.g. 2404:3100:1007:c789:851e:188f:1eb5:3b62, microsoft.com, or 5.188.10.0/24

81.18.87.130 CHECK

81.18.87.130 was not found in our database

ISP	TERRA SAT COMP ISP - Romania
Usage Type	Fixed Line ISP
ASN	Unknown
Hostname(s)	static-81-18-87-130.severin.rdsnet.ro
Domain Name	terrasat.ro
Country	 Romania
City	Giurgiu, Giurgiu County

Result:

- Not trusted mail servers are used , real company use secure mail servers.
- Trusted email providers don't use these Residential ISPs.
- Email says its from @ebay.com but domains of Hop's IPs are different here.

All above evidence shows this email is fake.

Task 8. Analyze the phishing email using automated tools:

1.Tool Used: MxToolbox (for header analyzer)

Phishing Email Sample : (already used in Q4)

Email Sender: aw-confirm@ebay.com

Headers Found

Header Name	Header Value
Return-Path	<aw-confirm@ebay.com>
X-Original-To	username@domain.com
Delivered-To	username@domain.com
Message-ID	<LCCOVYFMRBWQCHUDWYCOYUW@hotmail.com>
From	"aw-confirm@ebay.com" <aw-confirm@ebay.com>
Reply-To	"aw-confirm@ebay.com" <aw-confirm@ebay.com>
To	username@domain.com
Subject	Billing Issues
Date	Sun, 28 Nov 2004 00:30:27 +0400
MIME-Version	1.0
Content-Type	multipart/alternative; boundary="--591699981497957"

v=DMARC1; p=reject; rua=mailto:ebay@rua.agari.com; ruf=mailto:ebay@ruf.agari.com; fo=1; rf=afrf; pct=100

spf:ebay.com:81.18.87.130

Show

v=spf1 include:c._spf.ebay.com include:p._spf.ebay.com include:p2._spf.ebay.com ~all

Dkim Signature Error:

No DKIM-Signature header found - [more info](#)

Dkim Signature Error:

There must be at least one aligned DKIM-Signature for the message to be considered aligned. - [more info](#)

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	20.84.152.113	65.23.81.142		11/27/2004 8:24:27 PM	✓
2	9 minutes	unknown 81.18.87.130	sparky.domain.com	SMTP	11/27/2004 8:33:24 PM	✓

Header Analyzed

Email Subject: [Billing Issues](#)

Copy/Paste Warning

Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our [Email Deliverability tool](#)

Delivery Information

- DMARC Compliant
- SPF Alignment
- SPF Authenticated
- DKIM Alignment
- DKIM Authenticated

Results:

- No DKIM signature header found → authenticity cannot be verified.
- SPF Record:** SPF authenticated, alignment passed → but can be bypassed if DKIM/DMARC fails.
- DMARC Result:** Shows "compliant", but DKIM was missing and source domain is unknown, making this unreliable.

- IP 81.18.87.130 → residential/static IP in Romania (not a trusted eBay server).
- **Hostname:** spanky.dimain.com → not an official eBay mail server.
- Failed – mismatch between sender domain and actual sending domain.
- Only **2 hops**, and one is unknown → limited traceability indicates spoofing.

All above evidence shows this is phishing email.

2. Tool Used: AbuseIPDB

Phishing Email Sample: (already used in Q4)

Email Sender: aw-confirm@ebay.com

Verification of Hop's IPs

AbuseIPDB » 81.18.87.130

Check an IP Address, Domain Name, or Subnet
e.g. 2404:3100:1007:c789:851e:188f:1eb5:3b62, microsoft.com, or 5.188.10.0/24

81.18.87.130 CHECK

81.18.87.130 was not found in our database

ISP	TERRA SAT COMP ISP - Romania
Usage Type	Fixed Line ISP
ASN	Unknown
Hostname(s)	static-81-18-87-130.severin.rdsnet.ro
Domain Name	terrasat.ro
Country	Romania
City	Giurgiu, Giurgiu County

AbuseIPDB » 81.18.87.190

Check an IP Address, Domain Name, or Subnet
e.g. 2404:3100:1007:c789:851e:188f:1eb5:3b62, microsoft.com, or 5.188.10.0/24

81.18.87.190 CHECK

81.18.87.190 was not found in our database

ISP	RCS & RDS Residential
Usage Type	Fixed Line ISP
ASN	Unknown
Hostname(s)	static-81-18-87-190.severin.rdsnet.ro
Domain Name	digi.ro
Country	Romania
City	Giurgiu, Giurgiu County

Result:

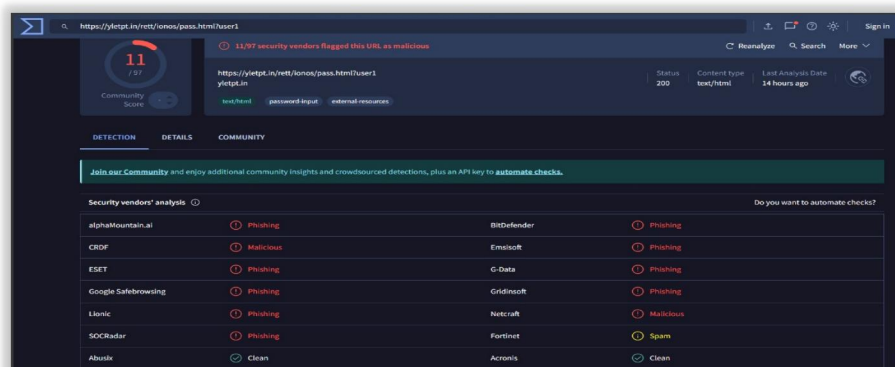
- Not trusted mail servers are used , real company use secure mail servers.
- Trusted email providers don't use these Residential ISPs.
- Email says its from @ebay.com but domains of Hop's IPs are different here.

All evidence shows this is phishing email.

1.Tool Used: VirusTotal

Purpose: Used for testing urls in emails whether they are harmful or not.

Url source: phishtank



Result:

- Scans indicates that 6 security engines flagged the url as phishing and malicious, suggesting a high risk and strong suspicion of malicious intent.